

(a) We can split this into

$$\text{total} = 0 \text{ numerals} + 1 \text{ numeral} + \text{at least 2 numerals}$$

$$\text{at least 2 numerals} = \text{total} - 0 \text{ numerals} - 1 \text{ numeral}$$

Let's start by finding the total. Since there are no restrictions, each position has 36 options

$$\text{total} = \underline{36} \cdot \underline{36} \cdot \underline{36} \cdot \underline{36} \cdot \underline{36} \cdot \underline{36} \cdot \underline{36} = 36^7$$

Now for 0 numerals, the restriction is that each character has to be an uppercase letter, so each position has 26 options

$$0 \text{ numerals} = \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} = 26^7$$

Lastly, 1 exact numeral means only one slot has 10 options, but out of 7 slots. So we multiply the count for 1 numeral by the ways we can put that one numeral. The rest of the slots MUST be letters

$$\begin{aligned} 1 \text{ numeral} &= \binom{7}{1} \cdot \underline{10} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} = 7 \cdot 10 \cdot 26^6 \\ &= 70 \cdot 26^6 \end{aligned}$$

Using our equation from above,

$$\text{at least 2 numerals} = \boxed{6^7 - 26^7 - 70 \cdot 26^6}$$

(b) First, let's find out the number of license plates with MILES being at the start

$$\underbrace{1}_M \cdot \underbrace{1}_I \cdot \underbrace{1}_L \cdot \underbrace{1}_E \cdot \underbrace{1}_S \cdot 36 \cdot 36 = 36^2$$

Now let's see how many times we can arrange MILES with 7 slots

1. MILESxx
2. xMILESx
3. xxMILES

Since there's three ways to putting MILES into 7 characters, we can just multiple 36^2 by 3 to get

$$\boxed{3 \cdot 36^2}$$

(c) We can think of this as:

1. choose 3 different characters out of 36
2. choose 3 different spots of 7 for the first character
3. choose 3 spots out of 4 for the second character

4. choose 1 spot out of 1 for the last character

Mathematically, this ends up being

$$\boxed{\binom{36}{3} \cdot \binom{7}{3} \cdot \binom{4}{3} \cdot \binom{1}{1}}$$

(d) Like the last one, we can think of this as

1. choose 3 different numerals out of 10
2. choose 1 letter out of 26
3. choose 3 spots out of 7 to place the numerals
4. permute the 3 numbers
5. choose 4 spots out of 4 to place the letters (basically just put the letters where the numbers aren't)

This ends up being

$$\begin{aligned} & \binom{10}{3} \cdot \binom{26}{1} \cdot \binom{7}{3} \cdot {}_3P_3 \cdot \binom{4}{4} \\ &= \boxed{\binom{10}{3} \cdot \binom{26}{1} \cdot \binom{7}{3} \cdot 3!} \end{aligned}$$

(e) For this one, since alphabetical order means A-Z (1 - 26) and there's 7 letters, the start can only be up to

$$26 - 7 + 1 = 20 \text{ (T)}$$

We can think of the first character essentially controlling the entire string, so in reality all we need to do is choose a valid first letter.

Using the range

$$[1, 20]$$

We have 20 letters to choose from, means our answer is just

$$\boxed{20}$$